MATH 241: Calculus III

Fall 2026

Lecture 19: Line Integrals

Background & Motivation

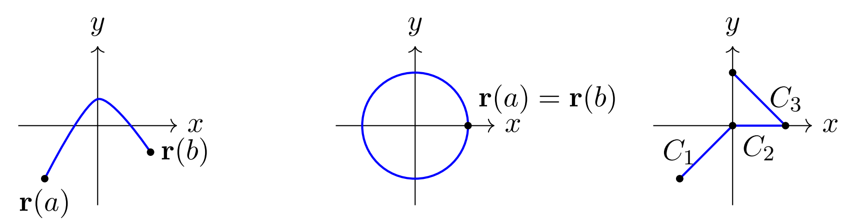
With vector fields defined, we now integrate them along curves. The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ computes the work done by a force along a path. For conservative fields, the Fundamental Theorem of Line Integrals gives us a shortcut: the integral depends only on the endpoints, not the path.

1. Contour Types

A **contour** (or **curve**) C is a path in $\mathbb{R}^2$ or $\mathbb{R}^3$ given by a parameterization $\mathbf{r}(t)$, $a \leq t \leq b$.

Definition 1: Types of Contours

- **Open** – endpoints $\mathbf{r}(a) \neq \mathbf{r}(b)$
- **Closed** – $\mathbf{r}(a) = \mathbf{r}(b)$ (loop); use notation $\oint_C$
- **Simple** – does not cross itself
- **Piecewise smooth** – finitely many smooth pieces joined end-to-end



Open Contour Closed Contour Piecewise Contour

Common Parameterizations

- Line segment from A to B – $\mathbf{r}(t) = (1-t)A + tB, \quad t \in [0, 1]$
- Circle $x^2 + y^2 = R^2$ – $\mathbf{r}(t) = \langle R\cos t,\, R\sin t\rangle, \quad t \in [0, 2\pi]$
- Graph $y = f(x)$, $a \leq x \leq b$ – $\mathbf{r}(t) = \langle t,\, f(t)\rangle, \quad t \in [a, b]$
- Helix – $\mathbf{r}(t) = \langle a\cos t,\, a\sin t,\, bt\rangle$

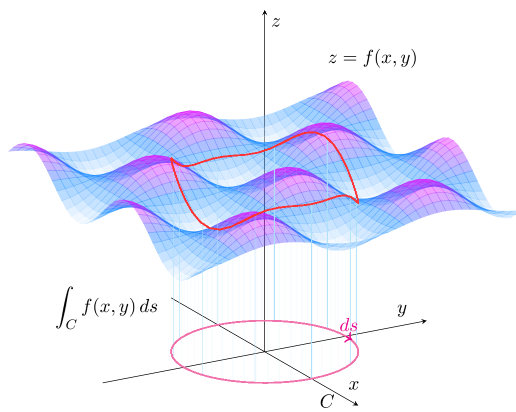
2. Scalar Line Integrals

Definition 2: Scalar Line Integral

For a scalar function f along a curve C parameterized by $\mathbf{r}(t)$:

$$\int_C f\,ds = \int_a^b f(\mathbf{r}(t))\,\|\mathbf{r}'(t)\|\,dt$$

– $ds = \|\mathbf{r}'(t)\|\,dt$ is the **arc length element**
– **Interpretation:** area of the "curtain" under f along C
– Direction does not matter: $\int_C f\,ds = \int_{-C} f\,ds$



Example 1: Scalar Line Integral Along a Line Segment

Evaluate $\displaystyle\int_C (x + 2y)\,ds$ where C is the segment from $(0, 0)$ to $(3, 4)$.

1. **Parameterize:** $\mathbf{r}(t) = \langle 3t,\, 4t\rangle$, $t \in [0, 1]$
2. **Speed:** $\mathbf{r}'(t) = \langle 3, 4\rangle$, $\quad \|\mathbf{r}'\| = \sqrt{9 + 16} = 5$
3. **Substitute:** $f(\mathbf{r}(t)) = 3t + 2(4t) = 11t$
4. **Integrate:** $\displaystyle\int_0^1 11t \cdot 5\,dt = 55\int_0^1 t\,dt = 55 \cdot \frac{1}{2} = \boxed{\frac{55}{2}}$

3. Vector Field Line Integrals

Definition 3: Line Integral of a Vector Field

For a vector field $\mathbf{F}$ along a curve C:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\,dt$$

– **Interpretation:** **work done by the force $\mathbf{F}$ along path C**
– Reversing direction **negates** the integral: $\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_C \mathbf{F} \cdot d\mathbf{r}$
– Equivalent notation: $\int_C P\,dx + Q\,dy + R\,dz$

Example 2: Work Around a Circle

Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle -y,\, x\rangle$ along the circle $x^2 + y^2 = 4$ (CCW).

1. **Parameterize:** $\mathbf{r}(t) = \langle 2\cos t,\, 2\sin t\rangle, \quad \mathbf{r}' = \langle -2\sin t,\, 2\cos t\rangle, \quad t \in [0, 2\pi]$
2. **Evaluate F:** $\mathbf{F}(\mathbf{r}(t)) = \langle -2\sin t,\, 2\cos t\rangle$
3. **Dot product:** $\mathbf{F} \cdot \mathbf{r}' = (-2\sin t)(-2\sin t) + (2\cos t)(2\cos t) = 4\sin^2 t + 4\cos^2 t = 4$
4. **Integrate:** $\int_0^{2\pi} 4\, dt = \boxed{8\pi}$

4. Fundamental Theorem of Line Integrals (FTOLI)

Theorem 1: Fundamental Theorem of Line Integrals

If $\mathbf{F} = \nabla f$ (conservative) and C goes from A to B:

$$\boxed{\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)}$$

Consequences:

- The integral depends only on **the endpoints** – **path independence**
- For any closed curve: $\oint_C \nabla f \cdot d\mathbf{r} = 0$
- If $\oint_C \mathbf{F} \cdot d\mathbf{r} \neq 0$ for some closed C, then $\mathbf{F}$ is **not** conservative

Example 3: Applying FTOLI

Evaluate $\int_C \langle ye^{xy},\, xe^{xy}\rangle \cdot d\mathbf{r}$ from $(0, 0)$ to $(1, 2)$.

1. **Recognize:** From Worksheet 21, $\mathbf{F} = \nabla f$ with $f(x, y) = e^{xy}$
2. **Apply FTOLI:** $\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 2) - f(0, 0) = e^{(1)(2)} - e^0 = \boxed{e^2 - 1}$

No parameterization needed — any path from $(0, 0)$ to $(1, 2)$ gives the same answer.

Example 4: Piecewise 3D Line Integral

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y^2,\, z^2,\, x^2\rangle$ along the path $(0, 0, 0) \to (1, 0, 0) \to (1, 1, 0) \to (1, 1, 1)$.

Segment 1: $(0, 0, 0) \to (1, 0, 0)$: $\quad \mathbf{r} = \langle t, 0, 0\rangle$, $\mathbf{r}' = \langle 1, 0, 0\rangle$
$\mathbf{F} = \left\langle 0, 0, t^2\right\rangle$, $\quad \mathbf{F} \cdot \mathbf{r}' = 0$. $\quad$ Integral $= 0$.
Segment 2: $(1, 0, 0) \to (1, 1, 0)$: $\quad \mathbf{r} = \langle 1, t, 0\rangle$, $\mathbf{r}' = \langle 0, 1, 0\rangle$
$\mathbf{F} = \left\langle t^2, 0, 1\right\rangle$, $\quad \mathbf{F} \cdot \mathbf{r}' = 0$. $\quad$ Integral $= 0$.
Segment 3: $(1, 1, 0) \to (1, 1, 1)$: $\quad \mathbf{r} = \langle 1, 1, t\rangle$, $\mathbf{r}' = \langle 0, 0, 1\rangle$
$\mathbf{F} = \left\langle 1, t^2, 1\right\rangle$, $\quad \mathbf{F} \cdot \mathbf{r}' = 1$. $\quad$ Integral $= \int_0^1 1\, dt = 1$.
Total: $0 + 0 + 1 = \boxed{1}$

Practice Problems

Problem 1. Evaluate $\int_C xy\, ds$ where C is the line segment from $(0,0)$ to $(1,1)$.

Answer: $\frac{\sqrt{2}}{3}$

Problem 2. Evaluate $\int_C \langle y^2,\, 2xy \rangle \cdot d\mathbf{r}$ from $(0,0)$ to $(1,1)$ along $y = x^2$.

Answer: 1

Problem 3. Evaluate $\oint_C \langle x^2,\, y^2 \rangle \cdot d\mathbf{r}$ where C is the unit circle. (*Hint: is the field conservative?*)

Answer: 0

Problem 4. Evaluate $\int_C \langle 2xz + y,\ x + z^2,\ x^2 + 2yz \rangle \cdot d\mathbf{r}$ from $(0,0,0)$ to $(1,2,3)$ using FTOLI.

(*Hint: find the potential function first.*)

Answer: $f = x^2 z + xy + yz^2; \quad f(1,2,3) - f(0,0,0) = 3 + 2 + 18 = 23$

Problem 5. Evaluate $\int_C (x^2 + y^2)\, ds$ where C is the helix $\mathbf{r}(t) = \langle \cos t,\, \sin t,\, t \rangle$, $0 \le t \le 2\pi$.

Answer: $2\sqrt{2}\,\pi$